12: Inference for Linear Combinations

Stat 230 - Spring 2026

> **Note**
>
> This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use https://aluby.github.io/stat230/activities/12-linear-combos

> **Note**
>
> I recommend finishing the Inference for MLR activity before starting. This activity will use the same data and it will be helpful to have worked through the equations for each category before starting!

1 Overview

The `cars` data set loaded below contains information on used cars for sale, including their `Price`, `Mileage`, and `Make` (a categorical variable with six levels).

> **Tip**
>
> Run the code chunks in order: objects you create in one chunk (like a fitted model) are available in the chunks that follow.

```
cars <- read.csv("https://aluby.github.io/stat230/data/Cars.csv", stringsAsFactors =
```

We'll proceed with the `car_lm2` model:

```
car_lm2 <- lm(log(Price) ~ Mileage * Make, data = cars)

summary(car_lm2)
```

2 Calculating CIs for linear combinations

To calculate a confidence interval for a linear combination of coefficients, we must first calculate the estimate and standard error of the linear combination.

Let's calculate a 95% confidence interval for the slope of the Cadillac model. Refer to your notes/slides and determine the formulas for the estimate and standard error of this slope.

Now that you know the formulas, you'll use R to implement them. First, extract the coefficients and covariance matrix from the model object. Try printing the results to see what they look like and make note of where the Cadillac coefficients are located.

```
# Extract the coefficients from the model object
car_coefs <- coef(car_lm2)

# Extract the covariance matrix from the model object
car_vcov <- vcov(car_lm2)

car_coefs
car_vcov
```

Next, calculate the estimate of the slope for Cadillacs.

```
estimate <- car_coefs[2] + car_coefs[8]
estimate
```

`car_coefs` is a vector, so we pull off the necessary coefficients using their position in square brackets. Here, `car_coefs[2]` pulls off the coefficient for `Mileage` and `car_coefs[8]` pulls off the coefficient for the interaction term `Mileage:MakeCadillac`.

Now, calculate the standard error of the slope for Cadillacs.

```
se <- sqrt(car_vcov[2,2] + car_vcov[8,8] + 2*car_vcov[2,8])
se
```

`car_vcov` is a matrix, so we pull off the necessary variances and covariances using their row and column positions in square brackets. Here, `car_vcov[2,2]` pulls off the variance for `Mileage`, `car_vcov[8,8]` pulls off the variance for the interaction term `Mileage:MakeCadillac`, and `car_vcov[2,8]` pulls off the covariance between these two coefficients.

Finally, use the estimate and standard error to calculate a 95% confidence interval for the slope of the Cadillac model. You can use the `qt()` function to find the appropriate critical value.

Q1. Calculate a 95% confidence interval for the slope of the Cadillac model.

```
# Fill in the blanks to calculate the 95% CI
# (hint: the degrees of freedom are in car_lm2$df.residual)
tstar <- qt(___, df = ___)

estimate - tstar * ___
estimate + tstar * ___
```

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3 Function quick reference

The following table summarizes the functions we used today:

Function	Purpose
`coef()`	Extract the estimated coefficients from a fitted model (as a named vector)
`vcov()`	Extract the covariance matrix of the estimated coefficients from a fitted model
`x[i]`, `m[i, j]`	Pull off element `i` of a vector, or the entry in row `i` and column `j` of a matrix
`sqrt()`	Take the square root (e.g., to turn a variance into a standard error)